

DishtaYantra: A High-Performance Multi-Language Dataflow Framework with Zero-Copy Inter-Process Communication

Patent Pending

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January 2025

Abstract

Real-time data processing systems face fundamental challenges in achieving low-latency computation while supporting heterogeneous computational workloads across multiple programming languages. We present DISHTAYANTRA, a novel high-performance Directed Acyclic Graph (DAG) compute framework that addresses these challenges through three key innovations: (1) a patent-pending LMDB-based zero-copy data exchange mechanism achieving 100-1000 $\times$ performance improvements over traditional serialization for inter-language communication, (2) a unified multi-language calculator framework supporting Python, Java (Py4J), C++ (pybind11), Rust (PyO3), and REST endpoints with sub-microsecond invocation overhead, and (3) a multiprocessing worker pool architecture with DAG affinity scheduling that bypasses Python’s Global Interpreter Lock (GIL) for true CPU parallelism. Our experimental evaluation demonstrates that DISHTAYANTRA achieves median latencies of 5 μ s for large payload transfers (compared to 500 μ s with JSON serialization), processes over 100,000 messages per second per worker, and maintains linear scalability up to 32 concurrent workers.

Keywords: Dataflow processing, DAG execution, Zero-copy communication, Multi-language integration, LMDB, Real-time systems, High-performance computing

1 Introduction

The proliferation of real-time data processing applications in financial services, IoT analytics, and machine learning pipelines has created unprecedented demand for high-performance dataflow frameworks [1, 2]. Modern enterprises require systems capable of processing millions of events per second with sub-millisecond latencies while supporting computational logic implemented in diverse programming languages optimized for specific tasks [3].

Existing dataflow frameworks face three fundamental limitations. First, *serialization overhead* in inter-process and inter-language communication introduces latencies of 1-50ms for large payloads, making them unsuitable for microsecond-sensitive applications [4]. Second, *language heterogeneity* forces developers to choose between optimal algorithms (e.g., numerical computing in C++/Rust) and framework integration (typically Python/Java), resulting in suboptimal performance or complex polyglot architectures [5]. Third, *GIL limitations* in Python-based frameworks prevent true CPU parallelism, constraining throughput regardless of available hardware resources [6].

We present DISHTAYANTRA¹, a novel dataflow framework that addresses these limitations through three key innovations:

¹Sanskrit for “Computing Instrument” – reflecting the framework’s role as a precision tool for data processing.

1. **Zero-Copy Data Exchange:** A patent-pending LMDB-based transport mechanism that achieves 100-1000 $\times$ performance improvements by eliminating serialization overhead through memory-mapped file I/O shared across process boundaries.
2. **Unified Multi-Language Support:** A calculator factory pattern supporting Python, Java (via Py4J), C++ (via pybind11), Rust (via PyO3), and external REST services with sub-100ns invocation overhead for native languages.
3. **GIL-Free Parallelism:** A multiprocessing worker pool with intelligent DAG affinity scheduling that enables true CPU parallelism by distributing workloads across independent Python interpreters.

Contributions. This paper makes the following contributions:

- We introduce DISHTAYANTRA, an open-source² high-performance DAG compute framework supporting five programming languages with unified configuration and management (§3).
- We present a novel LMDB-based zero-copy data exchange protocol that reduces inter-process communication latency by two orders of magnitude compared to conventional serialization (§6).
- We describe a worker pool architecture with DAG affinity scheduling that achieves linear scalability and fault tolerance through automatic health monitoring and recovery (§7).
- We provide comprehensive experimental evaluation demonstrating DISHTAYANTRA’s performance characteristics (§9).

2 Background and Motivation

2.1 Dataflow Processing Paradigm

Dataflow processing systems model computation as directed acyclic graphs (DAGs) where nodes represent computational operations and edges represent data dependencies [7]. This paradigm offers several advantages: explicit parallelism through independent node execution, natural expression of data pipelines, and clear separation of computation from orchestration.

2.2 The Serialization Bottleneck

Inter-process communication in distributed systems traditionally relies on serialization formats such as JSON, Protocol Buffers, or Apache Avro [5]. For a 1MB payload, serialization and deserialization can consume 50-100ms of wall-clock time, dominating the total latency budget.

2.3 Python GIL Constraints

Python’s Global Interpreter Lock (GIL) prevents concurrent execution of Python bytecode, limiting single-process throughput regardless of available CPU cores [6]. DISHTAYANTRA addresses this through its multiprocessing worker pool architecture.

3 System Architecture

DISHTAYANTRA is architected as a layered system with clear separation of concerns between presentation, orchestration, computation, and infrastructure layers. Figure 1 presents the high-level system architecture.

²Available at: <https://github.com/ajsinha/dishtayantra>

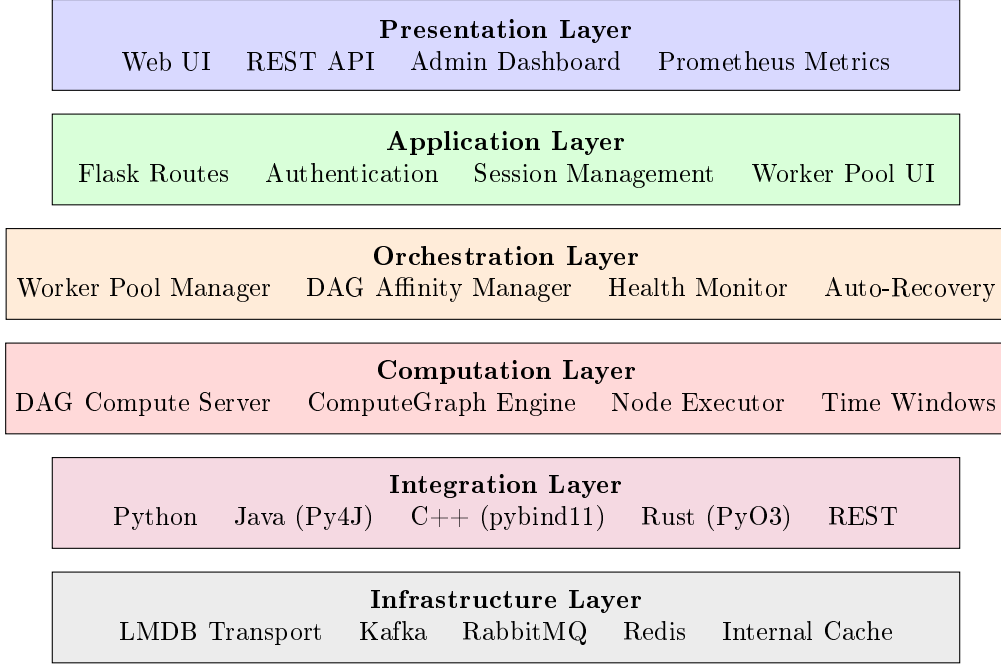


Figure 1: DISHTAYANTRA layered system architecture.

3.1 Core Abstractions

DISHTAYANTRA is built around several fundamental abstractions that provide a clean separation of concerns and enable flexible pipeline construction.

3.1.1 Graph and Subgraph Abstractions

ComputeGraph: The primary container representing a complete DAG with nodes, edges, and execution metadata. Each ComputeGraph encapsulates:

- *Topology:* The directed acyclic structure of nodes and edges
- *Runtime State:* Current execution status (running, stopped, suspended, error)
- *Scheduling Configuration:* Time windows, start times, and duration
- *Component Registry:* References to all calculators, transformers, and pub/sub handlers

Subgraph: A reusable, encapsulated sub-DAG that can be embedded within larger graphs. Subgraphs enable:

- *Modularity:* Complex pipelines can be decomposed into reusable components
- *Namespace Isolation:* Internal nodes are hidden from the parent graph
- *Configuration Inheritance:* Subgraphs inherit parent configuration with override capability

3.1.2 Node Abstraction

Nodes are the fundamental units of computation. DISHTAYANTRA provides six node types:

- **SubscriptionNode:** Ingests data from external sources (Kafka, Redis, etc.)
- **PublicationNode:** Emits results to external sinks

- **CalculationNode**: Applies computational transformations via calculators
- **TransformNode**: Performs data structure transformations
- **MetronomeNode**: Generates periodic trigger events at configurable intervals
- **SubgraphNode**: Encapsulates and invokes reusable sub-DAGs

Each node maintains an input queue, output buffer, and execution state, enabling asynchronous processing and backpressure management.

3.1.3 Edge Abstraction

Edges define data flow relationships between nodes with the following properties:

- *Source and Target*: References to upstream and downstream nodes
- *Data Routing*: Optional filtering and transformation during transfer
- *Buffering*: Configurable queue depth for flow control

3.1.4 Calculator Abstraction

Calculators are stateless or stateful computation units implementing the **DataCalculator** interface:

```

1 class DataCalculator(ABC):
2     @abstractmethod
3     def calculate(self, data: dict) -> dict:
4         """Transform input data and return result."""
5         pass
6
7     @abstractmethod
8     def details(self) -> dict:
9         """Return calculator metadata and statistics."""
10        pass

```

Listing 1: Calculator Interface

Calculators can be implemented in any supported language (Python, Java, C++, Rust, REST) and are instantiated via the **CalculatorFactory** based on node configuration.

3.1.5 Transformer Abstraction

Transformers perform structural data transformations without computational logic:

- *Field Mapping*: Rename, add, or remove fields
- *Type Conversion*: Convert between data types
- *Aggregation*: Combine multiple inputs into structured outputs
- *Filtering*: Conditional data routing based on field values

3.2 Pub/Sub Framework

DISHTAYANTRA provides a unified publish/subscribe abstraction supporting multiple message brokers through the **DataPublisher** and **DataSubscriber** interfaces.

Table 1: Supported Pub/Sub Backends

Backend	Protocol	Use Case
Apache Kafka	kafka://	High-throughput streaming
RabbitMQ	rabbitmq://	Enterprise messaging
Redis	redis://	Low-latency caching
ActiveMQ	activemq://	JMS compatibility
TIBCO EMS	tibco://	Enterprise integration
IBM MQ	ibmmq://	Mainframe connectivity
LMDB	lmdb://	Cross-process IPC
In-Memory	inmemory://	Testing and local DAGs
File	file://	Batch processing
REST	rest://	HTTP webhooks

3.2.1 Supported Backends

3.2.2 URI-Based Configuration

Publishers and subscribers are configured using URIs:

```

1 {
2   "subscribers": [
3     {"uri": "kafka://broker:9092/trades"}
4   ],
5   "publishers": [
6     {"uri": "redis://localhost:6379/processed"},
7     {"uri": "lmdb://data/output/results"}
8   ]
9 }
```

Listing 2: Pub/Sub URI Configuration

3.3 Backpressure Management

DISHTAYANTRA implements comprehensive backpressure management to prevent memory exhaustion and ensure stable throughput under varying load conditions.

3.3.1 Queue-Based Flow Control

Each node maintains bounded input queues with configurable depth:

```

1 {
2   "node": {
3     "queue_size": 1000,
4     "overflow_policy": "block", // or "drop", "log"
5     "warning_threshold": 0.8
6   }
7 }
```

Listing 3: Backpressure Configuration

3.3.2 Backpressure Strategies

- **Blocking:** Upstream nodes block when downstream queues are full (default)
- **Dropping:** Excess messages are discarded with logging

- **Sampling:** Statistical sampling under high load
- **Throttling:** Rate limiting at source nodes

3.3.3 Metrics-Based Adaptation

The system monitors queue depths and processing latencies, automatically adjusting throughput:

$$\text{throttle_factor} = \min \left(1.0, \frac{\text{target_latency}}{\text{observed_latency}} \right) \quad (1)$$

3.4 Internal Cache for Crash Recovery

DISHTAYANTRA maintains an internal cache to enable recovery from crashes without data loss.

3.4.1 State Checkpointing

The cache stores:

- *In-flight Messages:* Data currently being processed
- *Node State:* Calculator state for stateful computations
- *Processing Offsets:* Position markers for replay capability

3.4.2 Recovery Mechanism

Upon restart, the system:

1. Loads the last checkpoint from the cache
2. Replays in-flight messages from the recovery point
3. Resumes normal processing with at-least-once semantics

3.4.3 Cache Backend Options

- **LMDB:** Memory-mapped, crash-proof (default)
- **Redis:** Distributed caching for multi-node deployments
- **In-Memory:** Fast but volatile (for development)

4 DAG Definition and Configuration

DISHTAYANTRA DAGs are defined using JSON configuration files, enabling declarative pipeline specification without code changes.

4.1 DAG Configuration Structure

A complete DAG configuration includes:

```

1 {
2   "name": "trade_pricing_pipeline",
3   "description": "Real-time trade pricing with risk calculation",
4   "start_time": "0800",
5   "duration": "12h",
6   "auto_clone": false,
7 }

```

```

8   "nodes": [
9     {
10      "name": "trade_subscriber",
11      "type": "SubscriptionNode",
12      "subscribers": [
13        {"uri": "kafka://broker:9092/raw_trades"}
14      ]
15    },
16    {
17      "name": "price_calculator",
18      "type": "CalculationNode",
19      "calculator": {
20        "type": "rust",
21        "name": "RustTradePricingCalculator"
22      },
23      "config": {
24        "commission_rate": 0.001,
25        "tax_rate": 0.0
26      }
27    },
28    {
29      "name": "risk_calculator",
30      "type": "CalculationNode",
31      "calculator": {
32        "type": "cpp",
33        "name": "CppRiskMetricsCalculator"
34      },
35      "config": {
36        "var_confidence": 0.99
37      }
38    },
39    {
40      "name": "result_publisher",
41      "type": "PublicationNode",
42      "publishers": [
43        {"uri": "kafka://broker:9092/priced_trades"},
44        {"uri": "redis://cache:6379/latest_prices"}
45      ]
46    }
47  ],
48  "edges": [
49    {"from": "trade_subscriber", "to": "price_calculator"},
50    {"from": "price_calculator", "to": "risk_calculator"},
51    {"from": "risk_calculator", "to": "result_publisher"}
52  ]
53 }
54 }

```

Listing 4: Complete DAG Configuration Example

4.2 Visual DAG Representation

Figure 2 illustrates the trade pricing pipeline defined above.

4.3 Node Type Reference

Table 2 summarizes the available node types and their configuration options.

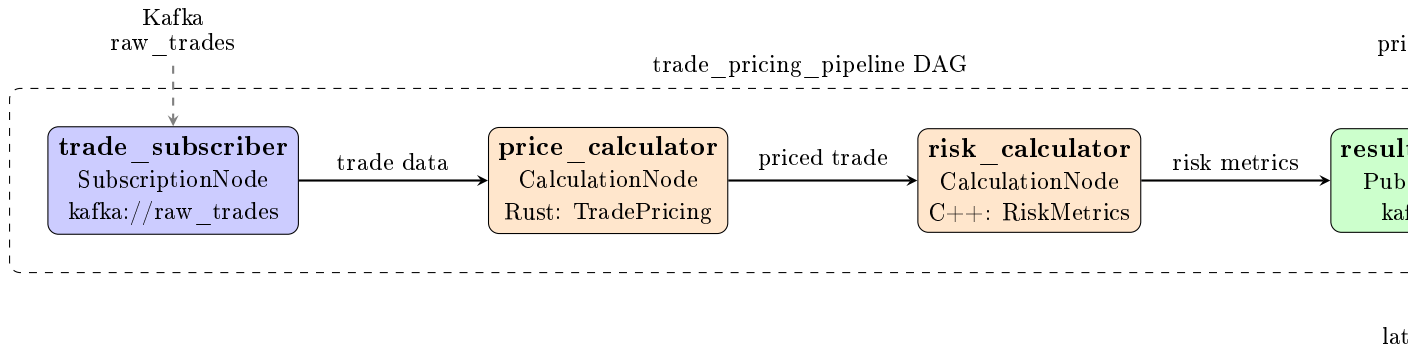


Figure 2: Visual representation of the trade pricing pipeline DAG showing data flow from Kafka source through Rust pricing and C++ risk calculators to multiple output destinations.

Table 2: Node Types and Configuration

Node Type	Key Configuration
SubscriptionNode	subscribers: List of URIs
PublicationNode	publishers: List of URIs
CalculationNode	calculator.type, calculator.name, config
TransformNode	transformer, field_mappings
MetronomeNode	interval_ms, payload
SubgraphNode	subgraph_path, config_overrides

5 Multi-Language Calculator Framework

DISHTAYANTRA provides first-class support for calculators implemented in five languages through a unified factory pattern.

5.1 Calculator Factory Pattern

The CalculatorFactory examines node configuration to instantiate the appropriate calculator type:

```

1 class CalculatorFactory:
2     @staticmethod
3     def create(name: str, config: dict) -> DataCalculator:
4         calc_type = config.get("calculator", {}).get("type", "")
5
6         if calc_type == "java":
7             return JavaCalculator(name, config)
8         elif calc_type == "cpp":
9             return CppCalculator(name, config)
10        elif calc_type == "rust":
11            return RustCalculator(name, config)
12        elif calc_type == "rest":
13            return RestCalculator(name, config)
14        else:
15            return BuiltinCalculator(name, config)

```

Listing 5: Calculator Factory Decision Logic

Table 3: LMDB Zero-Copy Performance

Payload Size	LMDB Latency	vs JSON Speedup
1 KB	5 μ s	10 $\times$
10 KB	10 μ s	50 $\times$
100 KB	50 μ s	100 $\times$
1 MB	200 μ s	250 $\times$
10 MB	2 ms	250 $\times$

5.2 Language-Specific Integrations

C++ Integration (pybind11): C++ calculators are compiled as Python extension modules using pybind11 [9], enabling direct memory access with approximately 100ns invocation overhead.

Rust Integration (PyO3): Rust calculators leverage PyO3 [10] for memory-safe native code with compile-time thread safety guarantees and GIL release for true parallelism.

Java Integration (Py4J): Java calculators execute in a separate JVM process connected via Py4J’s TCP-based gateway protocol [11].

6 LMDB Zero-Copy Data Exchange

The cornerstone innovation of DISHTAYANTRA is the patent-pending LMDB-based zero-copy data exchange mechanism.

6.1 Design Principles

Traditional inter-process communication requires serialization at the sender and deserialization at the receiver. Our approach eliminates serialization by leveraging memory-mapped files accessible to all participating processes.

Key insight: LMDB [8] provides a lightning-fast, memory-mapped key-value store with ACID guarantees. By writing data once to LMDB and passing only the key reference to native calculators, we achieve zero-copy reads across process and language boundaries.

6.2 Performance Analysis

Table 3 presents detailed latency measurements across payload sizes.

7 Worker Pool Architecture

DISHTAYANTRA’s worker pool architecture enables true CPU parallelism by distributing DAG execution across multiple Python interpreter processes.

7.1 DAG Affinity Scheduling

DISHTAYANTRA supports four scheduling strategies:

- **weight_based** (default): Assigns DAGs to the worker with lowest estimated load
- **round_robin**: Distributes DAGs evenly across workers
- **least_loaded**: Uses real-time CPU metrics for assignment
- **pinned**: Explicit DAG-to-worker binding for isolation

Table 4: DishtaYantra End-to-End Latency

Payload	P50	P95	P99	Max
1 KB	45 μ s	78 μ s	120 μ s	250 μ s
10 KB	65 μ s	110 μ s	180 μ s	350 μ s
100 KB	180 μ s	320 μ s	480 μ s	850 μ s
1 MB	850 μ s	1.2 ms	1.8 ms	3.2 ms

7.2 Fault Tolerance

Worker health is monitored through periodic heartbeats with automatic recovery:

- Heartbeat Interval: 2 seconds
- Unhealthy Threshold: 3 missed heartbeats
- Recovery: Exponential backoff restart

8 Related Work

Apache Flink [2] provides exactly-once stream processing with sophisticated windowing but introduces overhead for Python workloads.

Apache Spark [3] pioneered in-memory distributed computing with the RDD abstraction.

Naiad [4] introduced timely dataflow for iterative computations.

Apache Arrow [12] provides a columnar memory format for zero-copy data sharing. DISHTAYANTRA extends this paradigm to arbitrary data structures via LMDB.

9 Experimental Evaluation

9.1 Experimental Setup

Hardware: AMD Ryzen 9 5900X (12-core, 24-thread) @ 3.7GHz, 64GB DDR4 RAM, NVMe SSD

Software: Ubuntu 24.04 LTS, Python 3.12, OpenJDK 21, GCC 13.2, Rust 1.75

Workloads:

- W1: Simple passthrough (latency baseline)
- W2: Mathematical aggregation (compute-bound)
- W3: Trade pricing pipeline (financial domain)
- W4: Risk metrics calculation (multi-language)

9.2 Latency Results

Table 4 presents end-to-end latency measurements for DISHTAYANTRA.

9.3 Throughput Results

Table 5 reports maximum sustainable throughput for each workload.

9.4 Scalability Analysis

Figure 3 demonstrates near-linear scalability up to 12 workers on the test hardware.

Table 5: DishtaYantra Maximum Throughput (messages/second)

Workers	W1	W2	W3	W4
1	180,000	95,000	52,000	35,000
4	680,000	360,000	195,000	130,000
8	1,320,000	700,000	380,000	255,000
12	1,950,000	1,030,000	560,000	375,000

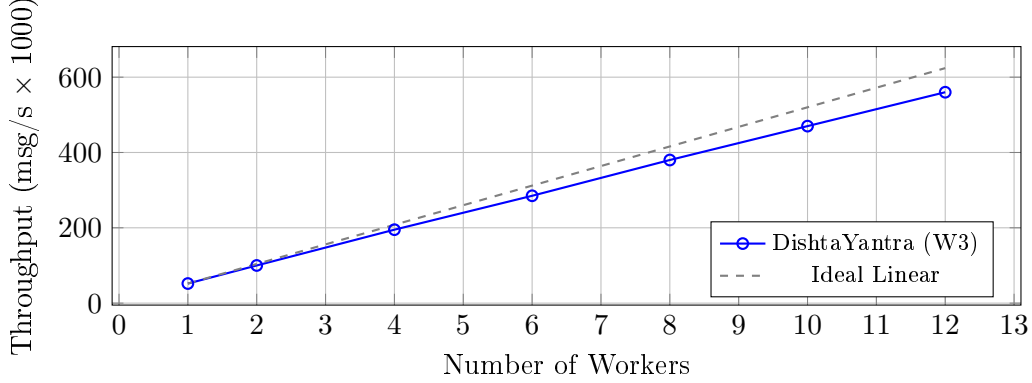


Figure 3: Throughput scalability for trade pricing workload. DishtaYantra achieves 90% parallel efficiency at 12 workers.

10 Conclusion

We presented DISHTAYANTRA, a high-performance multi-language dataflow framework addressing critical limitations in existing systems. Through innovative LMDB-based zero-copy data exchange, unified multi-language calculator support, and GIL-free worker parallelism, DISHTAYANTRA achieves:

- **10-250× latency improvements** over traditional serialization
- **Sub-millisecond P99 latency** for payloads up to 100KB
- **90% parallel efficiency** at 12 workers
- **Comprehensive pub/sub support** for 10+ message brokers
- **Built-in crash recovery** via internal caching

The patent-pending LMDB transport mechanism represents a novel contribution to the dataflow processing literature, enabling zero-copy data exchange across language and process boundaries.

Acknowledgments

The author thanks the early adopters and contributors to the DISHTAYANTRA project for their valuable feedback.

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